

How DNS Works (And How My Custom Domain Will Connect)

To understand how my portfolio will eventually be served at `melihyilmaz.flyrank.ai`, we first need to understand DNS (Domain Name System). Think of DNS as the internet's phonebook. Humans remember names (like `flyrank.ai`), but servers communicate using numbers (IP addresses). DNS translates the human-friendly names into computer-friendly numbers.

The Journey of a Request When someone types my URL into their browser, a rapid background process occurs:

- **Resolver:** The browser asks a recursive DNS resolver (usually provided by the user's Internet Service Provider) if it knows the target IP address. If it doesn't have the answer cached, the resolver starts hunting.
- **Nameserver:** The resolver queries the Root server, then the TLD (Top-Level Domain) server, and finally reaches the authoritative **Nameserver**. This final nameserver holds the actual, definitive truth about my specific domain.
- **Record:** The nameserver checks its zone file for the specific DNS **Record** associated with the requested URL.
- **Response:** The nameserver sends this record back to the resolver, which passes it to the browser. The browser now knows exactly where to connect and securely loads the website.

The CNAME Record and My Configuration For my capstone, I won't be pointing my FlyRank subdomain directly to a static IP address, because modern hosting platforms like Vercel use dynamic IP addresses that can change. Instead, I will use a **CNAME (Canonical Name)** record.

A CNAME record acts as an alias. It maps one domain name to another domain name rather than an IP.

When my subdomain is provisioned, the DNS record will be configured exactly like this:

- **Record Type:** `CNAME`
- **Name/Host:** `melihyilmaz` (which resolves to `melihyilmaz.flyrank.ai`)
- **Value (Target):** `melihyilmaz.vercel.app`

By doing this, the DNS system is instructed: "When someone looks for `melihyilmaz.flyrank.ai`, route them to `melihyilmaz.vercel.app`." Vercel's servers will recognize the incoming request for my custom domain, automatically provision the SSL/TLS certificate for the padlock icon, and serve my HTML files securely over HTTPS.